

Momentum Episodes

How the system detects, tracks, and notifies significant price moves — explained for leadership

Primary example ticker: SNDK · Multi-page deep dive with charts & tables

Purpose of this document

This brief explains the **Momentum Episodes** product end-to-end so leadership can understand what an “episode” is, when users get phone pushes, what stays silent, how thresholds work, and how a real trading day on **SNDK** would look. Numbers below use the **default equity configuration** shipping in code (all tunable via env / settings UI).

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1. Executive summary

Traders and news desks do not need a push every time a stock ticks. They need a **coherent story of a move**: when it started, whether it is still extending, whether it is fading, and whether it has fully reversed. That story is an **Episode**.

Our backend polls market data (Yahoo 1-minute candles by default every ~60 seconds), computes percentage moves over many lookback windows (5 minutes, 1 hour, day vs prior close, etc.), and when a window **freshly crosses** its absolute threshold, it opens one ACTIVE episode for that ticker. While the episode is alive, internal states update often (HOLDING, WEAKENING, ...) but the phone only rings on a short list of material events.

In one sentence

Definition

An **Episode** is a single directional momentum event for one ticker: it freezes a reference price and window at START, tracks peak/trough and giveback until the move expires, reverses, or the equity cash session closes — and it pushes only when the story meaningfully changes.

Why this exists

- **Reduce alert spam** — one coherent episode instead of dozens of overlapping “% in 5m / 15m / 1h” pings.
- **Preserve narrative** — “started at +5%, later +8% (acceleration), then major fade, then reversed”.
- **Mobile-first copy** — titles look like **[green] SNDK +5.3% in last 5 minutes (UP)** or **[red] ... (DOWN)**, with research-backed body text on start.
- **Multi-asset ready** — equity defaults shown here; commodities, crypto, forex, indexes have separate threshold maps.

What leadership should remember

Topic	Rule of thumb
One ACTIVE episode	At most one open episode per ticker at a time
Who starts it	Shortest lookback window that freshly crosses its threshold
Phone pushes	START, +2pp acceleration, strong fade (≥60% giveback), re-accel, reversal
Usually silent	HOLDING, mild WEAKENING, inactivity expiry, equity market close
Day vs rolling	“day” = vs Yahoo previous regular close (not a rolling N-hour window)
Default poll	~60 seconds per ticker (configurable)

2. System overview

The engine is multi-ticker. **SNDK** is the primary / bootstrap symbol in configuration, but the same logic applies to any watched equity (and other asset classes with their own thresholds). A background loop fetches candles, computes returns, advances the episode state machine, persists history, and routes push-worthy events to Expo.

How the System Runs (High Level)

Poll ~every 60s for each watched ticker (default primary: SNDK)

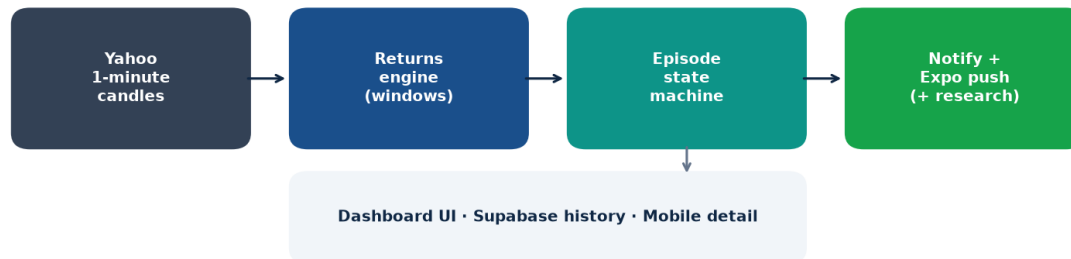


Figure 1 — High-level data path from market data to push and dashboard.

Main building blocks

Component	Responsibility
Candles / Yahoo	1-minute (and daily) bars; previous close for the “day” return
Returns engine	Move % for each lookback window; exact elapsed span labels
Detector	Which windows are above threshold; shortest-window START pick; re-arm checks
Episode state machine	Create / update / expire / reverse one ACTIVE episode per ticker
Notify copy	Human titles/bodies; hides backend jargon from the user
Auto-start research	Optional research “likely driver” for START push body
Delivery	Expo push, idempotency keys, dry-run / allowlist modes
Store + Supabase	Active episode, event ring buffer, thresholds, digests
Dashboard / mobile	Live returns grid, episode panel, notification history
Daily digest	OPEN / MIDDAY / CLOSE summaries — separate from episodes

Polling model

Default poll interval is **60 seconds** (MOMENTUM_POLL_MS). Within a cycle the engine walks the watchlist with a small gap between symbols to stay rate-limit friendly. On each tick for a symbol it recomputes all returns, runs `advanceEpisode()`, emits zero or more events, and updates the active episode snapshot.

3. Rolling returns & thresholds

Every poll produces a map of **returns**: for each window key, the percent change from a historical anchor price to the latest print. Absolute move matters — both UP and DOWN can open episodes.

How a return is calculated

Rolling window (e.g. 5m, 3h, 24h): find the candle at or before (now – window), then

$$\text{move\%} = (\text{currentPrice} - \text{historicalPrice}) / \text{historicalPrice} \times 100$$

day (special): percent change vs Yahoo's **previous regular-session close**, not “last 24 hours of candles”. This is the classic “today’s session move” number users expect on equities.

Gaps (overnight / weekend): equities have no 1m bars while closed. Longer windows that land in a gap anchor to the last print before the hole (often Friday), so Monday open can still show multi-hour moves. Bridge windows (30h / 40h / 50h) exist mainly for Monday pre-open UI context; by default they do **not** start episodes.

Default equity thresholds (episode-capable)

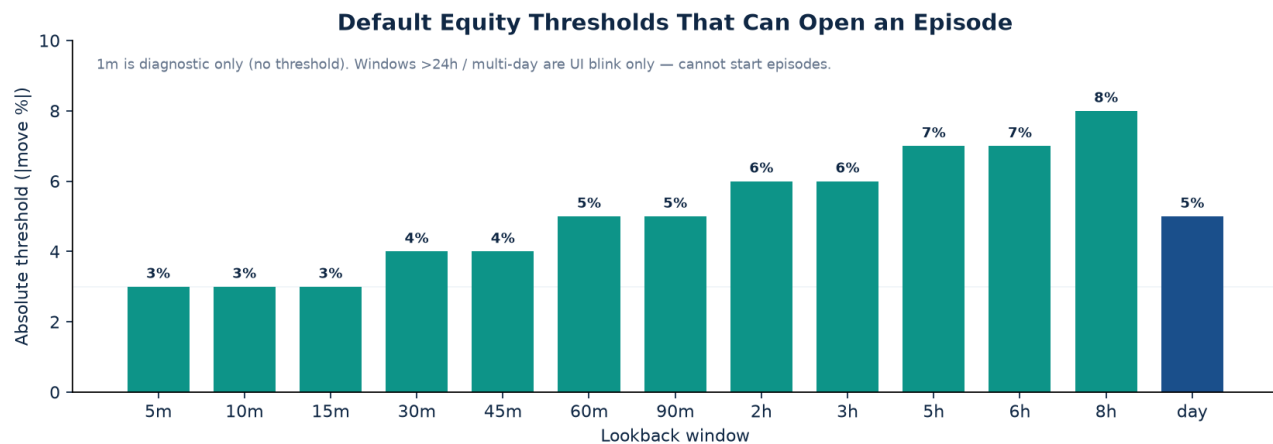


Figure 2 — Default absolute % thresholds. A window “hits” when $|\text{move\%}| \geq \text{threshold}$.

Window	Default thr	Can start episode?	Notes
1m	— (null)	No	Diagnostic only
5m / 10m / 15m	3%	Yes	Fast tape / breakout style
30m / 45m	4%	Yes	
60m / 90m	5%	Yes	
2h / 3h	6%	Yes	
5h / 6h	7%	Yes	
8h	8%	Yes	
10h–24h	off by default	Only if thr set	Extended; often diagnostic
day	5%	Yes	Vs previous close
1w / 10d / ... / 1y	off	No	UI blink / research context only
30h / 40h / 50h	off	No	Weekend bridge cards (UI)

Important product rule: **only ≤24h rolling windows and day** are episode-eligible. Longer horizons can light up the dashboard but will not open or keep an episode alive. That keeps episodes about “what’s happening now / today,” not “the stock is up a lot this quarter.”

Asset classes

Thresholds are stored per class: equity, commodity, forex, crypto, index. Editing equity 5m does not change crypto 5m. Values persist to disk and Supabase so restarts keep operator tuning.

4. What is an Episode

Think of an episode as a **case file** the system opens when a ticker's move first becomes “worth tracking,” then updates until that case is closed. It is not a trade recommendation; it is a structured momentum narrative with frozen identity fields and live measurements.

Frozen at START (identity)

Field	Meaning
episodeId	Unique id (ticker + time + random) for history / idempotent pushes
direction	UP or DOWN — fixed for the life of the episode
detectedWindow / windowType	Which lookback won START (e.g. 5m) — never re-bound later
referencePrice / referenceTime	Anchor price & time of that window at trigger
triggerPrice / triggerTime	Price & time when START fired
initial / trigger move %	Immutable size of the move at open
exactMinutes / exactLabel	True elapsed span (e.g. 48 minutes), not only the bucket name

Live fields (update every poll)

Field	Meaning
currentPrice / currentMove%	Live price and % vs frozen reference
peak / trough	Best extreme in the episode direction
peakMovePercent	Largest move reached
giveback ratio	How much of the peak move has been given back
state	STARTED, HOLDING, ACCELERATING, WEAKENING, ...
status	ACTIVE, EXPIRED, REVERSED, CLOSED_AT_MARKET_CLOSE, ENDED
lastNotifiedEpisodeMovePct	Last move % we actually pushed about (for +2pp rule)
meaningfulExtreme + lastMaterialProgressAt	Inactivity clock — resets on ≥0.5pp new extreme

START selection rule (shortest window wins)

On a given poll, many windows can all be above threshold (e.g. 5m +3.4%, 3h +7.2%, day +6%). We do **not** start from the strongest move. We start from the **shortest qualifying eligible window** among **fresh threshold crosses**. That freezes a tight narrative (“in last 5 minutes”) instead of a stale long window.

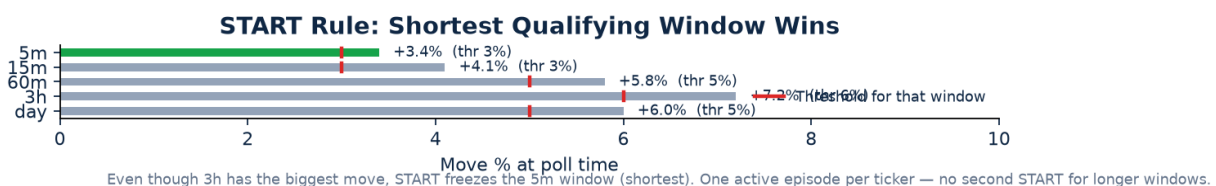


Figure 3 — Multiple windows qualify; 5m wins START even though 3h has a larger %. Red marks show each window's threshold.

Fresh cross requirement

A START only happens when a window **newly** goes from below → at/above threshold (edge detection across polls). If the process restarts and 3h is already +7%, edges are seeded first — no retroactive START spam. Windows that stay elevated must cool and re-cross to start again after a full re-arm (see §8).

One active episode per ticker

While ACTIVE, a newly qualifying shorter window does **not** open a second START. The existing case continues; acceleration is measured against the frozen reference and last notified move, not by spawning parallel stories.

5. Lifecycle states

Episode Lifecycle (Backend States)

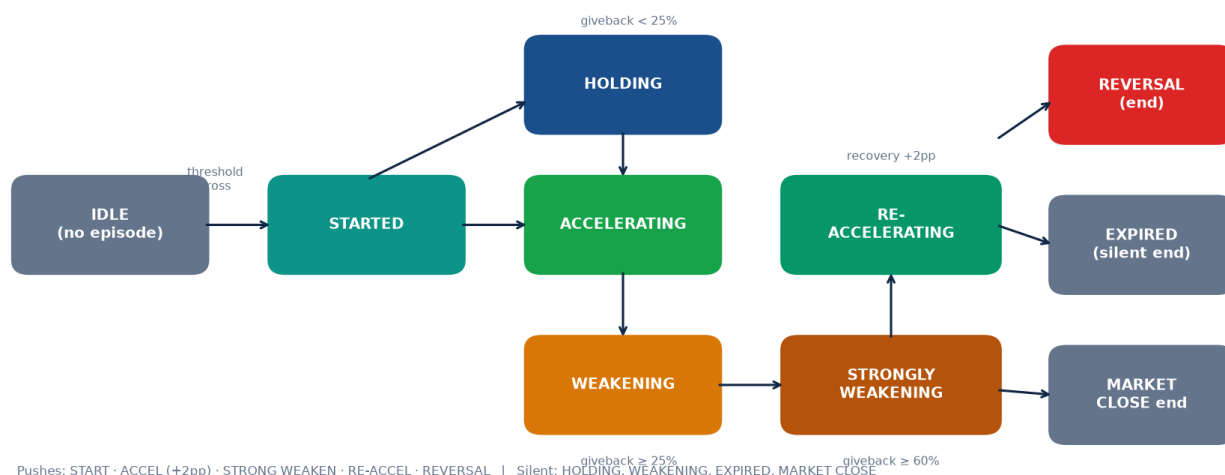


Figure 4 — Backend state graph. Terminal boxes end the episode; only some transitions push.

State dictionary

State	What it means	Typical push?
STARTED	Just opened; dwell briefly before silent reclassify to HOLDING	Yes (async research path)
ACCELERATING	Move extended $\geq +2pp$ beyond last notified episode move	Yes
HOLDING	Move intact; giveback under $\sim 25\%$ of peak	No
WEAKENING	Giveback $\geq \sim 25\%$ (hysteresis back only under $\sim 20\%$)	No
STRONGLY_WEAKENING	Giveback $\geq \sim 60\%$ of peak move	Yes (once per fade cycle)
RE_ACCELERATING	After strong fade, recovered $+2pp$ from recovery anchor	Yes
REVERSAL	Original move erased AND opposite direction threshold hit	Yes (ends episode)
EXPIRED	No meaningful progress for 60 minutes (default)	No
CLOSED_AT_MARKET_CLOSE	US equity cash session closed — archive	No

Statuses vs states

status is the case lifecycle (ACTIVE vs terminal). **state** is the qualitative phase while ACTIVE (or the terminal reason label). Analytics and mobile detail screens use both.

6. Notification policy (push vs silent)

This is the most important product distinction. The backend is chatty for correctness; the phone is selective so users trust alerts. Operators reviewing logs will see many MOMENTUM_STATE / ENDED rows that never left the server.

What Users Actually Receive on Their Phone

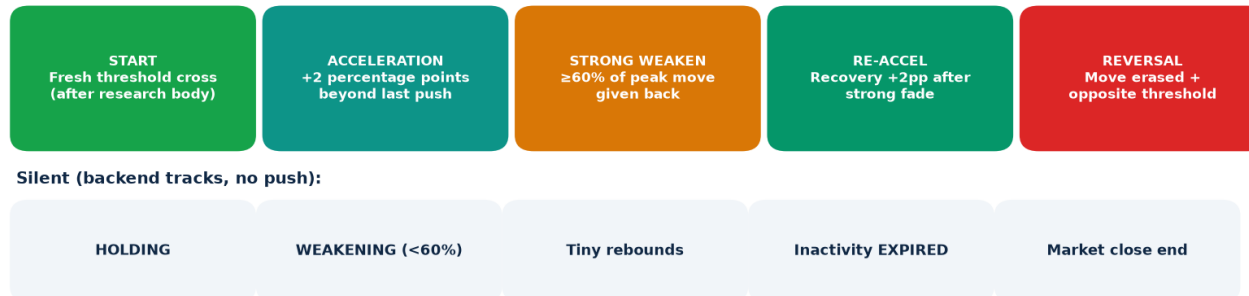


Figure 5 — User-visible pushes vs silent backend tracking.

Push-worthy event types

Event type	When	Example title pattern
MOMENTUM_STARTED	Fresh thr cross + research (or fallback body)	[green] SNDK +5.3% in last 5 minutes
MOMENTUM_ACCELERATING	Episode move ≥ last notified + 2pp	SNDK gains another +2.1% in 18 mins
MOMENTUM_STRONG_WEAKENING	≥60% giveback of peak; once per cycle	SNDK is giving back the move
RE_ACCEL (accel subtype)	After strong fade, +2pp recovery	SNDK is accelerating again
MOMENTUM_REVERSED	Move erased + opposite thr hit	SNDK reversed — now -3.1% ...

Silent by design

- HOLDING / mild WEAKENING state labels (dashboard only)
- Inactivity EXPIRED after no ≥0.5pp new extreme for 60 minutes
- Equity MARKET_CLOSE archive at regular-session close
- Sub-threshold wiggles and sub-2pp extensions
- Long-horizon “blink only” window flashes

START push timing note

START is intentionally special: the event is recorded, then an async research path may fetch a “likely driver” and send the Expo push. A max-age guard (default 5 minutes) suppresses a START push that would arrive late (“+3.5% in 5 min” half an hour later is worse than silence).

Idempotency

Pushes carry keys of the form episodeld:eventType:cycle so retries or dual workers do not double-notify. Strong-weakening and acceleration maintain cycle counters so a later fade after a re-acceleration can alert again.

7. Acceleration, giveback, fade & reversal

7.1 Acceleration (+2 percentage points)

After START (or any prior push), we remember **lastNotifiedEpisodeMovePct**. Another acceleration push fires only when the live episode move extends by at least **2.0 percentage points** beyond that last notified level (ACCELERATION_ALERT_DELTA_PP). Example: started and notified at +5.3% → next push needs about +7.3% episode move or more.

This is **not** “price moved 2% since last alert” in isolation — it is measured on the episode move vs the frozen reference, which keeps the narrative consistent when the window is 5m but the episode has been open for an hour.

7.2 Giveback math

For an UP episode: gain = peakPrice – referencePrice; giveback ratio = (peakPrice – currentPrice) / gain.

For DOWN: drop = reference – trough; giveback = (current – trough) / drop.

Defaults: HOLDING if giveback < 25%; WEAKENING at ≥25%; return to HOLDING only after ≤20% (hysteresis); STRONGLY_WEAKENING at ≥60%.

Giveback of peak move	Backend state	Push?
0% – 25%	HOLDING (or STARTED early on)	No (except START itself)
25% – 60%	WEAKENING	No
≥ 60%	STRONGLY_WEAKENING	Yes — once per weakening cycle
Then recovers +2pp from fade anchor	RE_ACCELERATING	Yes

7.3 Strong fade is not reversal

Product language is careful: a 60% giveback means the move is **fading hard**, not that the thesis inverted. True **REVERSAL** requires (1) the original directional move is erased (price back through reference), and (2) an opposite-direction eligible window is now through its threshold. Only then do we push REVERSED and end the episode.

7.4 Material progress (silent clock)

Separately from pushes, if the extreme extends by ≥ **0.5 pp**, we refresh lastMaterialProgressAt. If no such progress for **60 minutes**, the episode EXPIRES silently and the direction goes through full re-arm policy.

8. Ending, re-arm, and market close

How episodes end

End reason	Trigger	Push?	Re-arm policy
REVERSAL	Move erased + opposite thr	Yes	FULL cool-off
INACTIVITY	No ≥ 0.5 pp progress for 60m	No	FULL cool-off
MARKET_CLOSE	US equity/index session CLOSED	No	SESSION (lighter)
OUT_OF_SCOPE	Window not ≤ 24 h / day	No	FULL
Manual end (ops)	Operator / API	No	Often wait for quiet

FULL re-arm (after expire / reverse)

After a full end, the same direction is **disarmed** until **every** same-direction eligible window cools below its own (threshold – 1pp buffer). Example: 5m thr 3% must fall under 2%; 3h thr 6% under 5%. This stops the system from immediately re-opening the same stretched move as a “new” episode.

SESSION policy (market close)

Equities and indexes: when the regular US cash session closes, any ACTIVE episode is archived as CLOSED_AT_MARKET_CLOSE **without** a phone push and without the full multi-window cool-off. Tomorrow’s genuine fresh 5m jump can START even if the 24h return is still elevated — but START still requires a **threshold crossing** edge, not “already hot on restart.” Crypto / forex / many commodities do not hard-close on US equity hours; they rely on inactivity expiry.

Cold start / deploy safety

On first observation after process start, threshold edge state is seeded. Already-elevated windows do not fire a barrage of STARTs for moves that began hours earlier. This is deliberate ops hygiene.

9. Full illustrative trading day — SNDK

How to read this example

Prices below are **illustrative** (synthetic path shaped like a real momentum day) so managers can see every major phase in one session. The rules, thresholds, push copy patterns, and state transitions match production logic. Assume previous regular close = **\$100.00**, default equity thresholds, poll ≈ 60 s, ET regular hours 9:30–16:00.

9.1 Price path

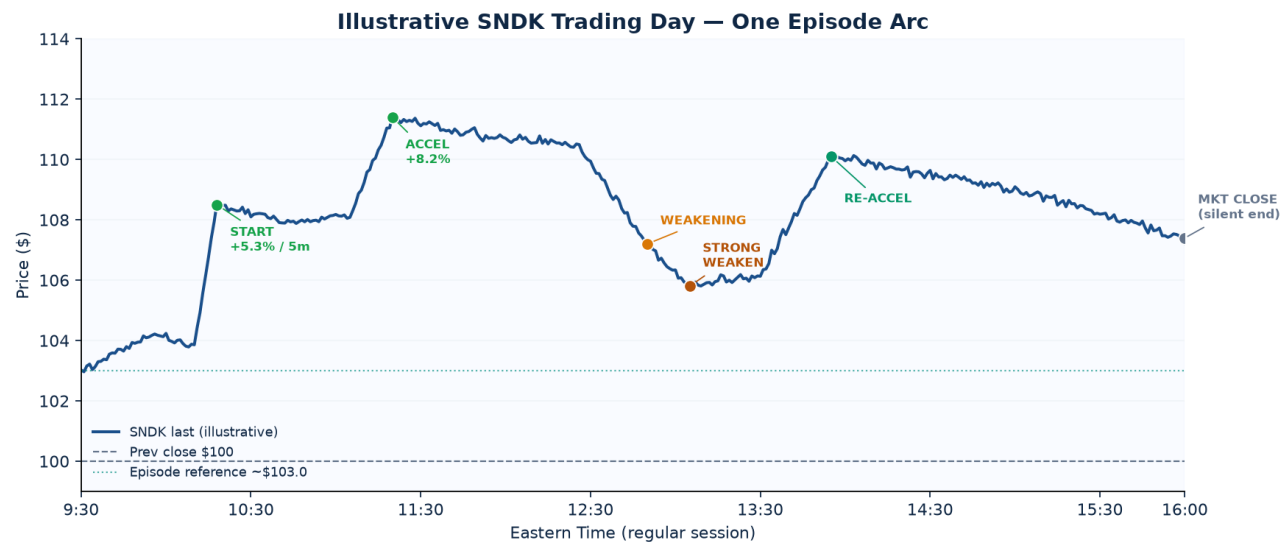


Figure 6 — Illustrative SNDK regular-session price with episode events marked.

9.2 Episode move & giveback

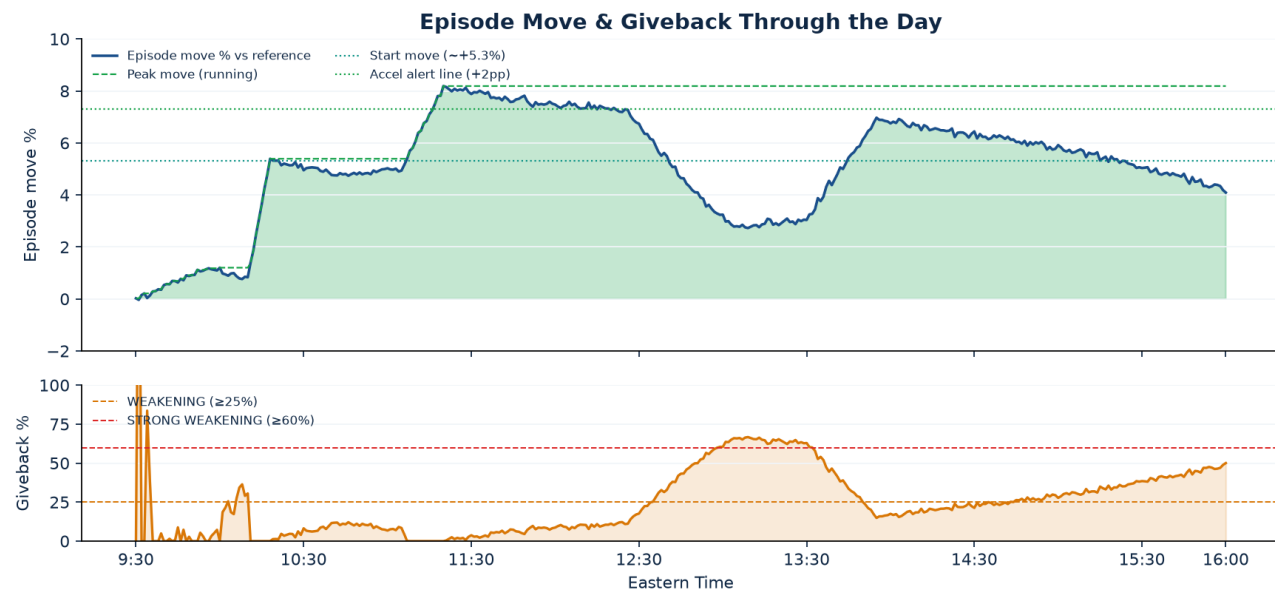


Figure 7 — Episode move % vs frozen reference (~\$103) and giveback of peak move through the day.

9.3 Minute-by-minute narrative (condensed)

ET time	Price	What the system does	User sees
09:30	\$103.0	Gap open; day $\approx$ +3% vs \$100 — below day thr 5%. Edges seed if cold start. No episode yet.	Dashboard returns only
09:55	\$104.2	Grind higher; short windows still under 3%.	Quiet
10:18	\$108.5	5m return freshly crosses +3% thr (shortest among hits). OPEN episode UP. Freeze ref=\$103.0, window=5m, move $\approx$ +5.3%. State STARTED.	Push START after research
10:20–10:45	~\$108	Dwell then silent $\rightarrow$ HOLDING. Peak updates slightly. No +2pp yet vs last notified +5.3%.	No new push
11:20	\$111.4	Episode move $\approx$ +8.2% ($\geq$ 5.3+2). Emit ACCELERATING. Update lastNotified to +8.2%. Peak=\$111.4.	Push ACCEL
11:20–12:30	~\$110.5	HOLDING; giveback small. Material progress clock refreshed near peak.	Quiet
12:50	\$107.2	Giveback $\approx$ 50% of peak gain $\rightarrow$ WEAKENING (still silent).	Dashboard state only
13:05	\$105.8	Giveback $\geq$ 60% $\rightarrow$ STRONGLY_WEAKENING. One strong-fade push this cycle. Recovery anchor set.	Push STRONG FADE
13:55	\$110.1	Recovery from fade anchor $\geq$ +2pp $\rightarrow$ RE_ACCELERATING push. Cycle flags reset for a future fade.	Push RE-ACCEL
14:00–15:40	~\$109	HOLDING / mild chop. No further +2pp beyond last notified recovery level.	Quiet
16:00	\$107.4	US cash CLOSED $\rightarrow$ CLOSED_AT_MARKET_CLOSE. Episode archived. SESSION re-arm (no full cool-off).	No push

9.4 Sample notification copy (what the manager would see on a phone)

#	Type	Example title	Example body
1	START	[green] SNDK +5.3% in last 5 minutes	Likely driver from research (e.g. sector bid / headline). Fallback: "Sharp upward momentum detected."
2	ACCEL	SNDK gains another +2.9% in 62 mins	Total episode move now about +8.2% from the frozen reference; elapsed since last alert called out in body patterns.
3	STRONG FADE	SNDK is giving back the move	Peak extension has faded $\geq$ 60%; not a full reversal — price still above reference.
4	RE-ACCEL	SNDK is accelerating again	The move recovered from the post-fade level back toward / beyond the prior extension by $\geq$ 2pp.

9.5 Numbers at key checkpoints

Checkpoint	Ref \$	Price \$	Episode move	Peak move	Giveback	State
START 10:18	103.00	108.50	+5.3%	+5.3%	0%	STARTED
ACCEL 11:20	103.00	111.40	+8.2%	+8.2%	0%	ACCELERATING
HOLD mid-day	103.00	110.50	+7.3%	+8.2%	~11%	HOLDING
WEAKEN 12:50	103.00	107.20	+4.1%	+8.2%	~50%	WEAKENING
STRONG 13:05	103.00	105.80	+2.7%	+8.2%	~67%	STRONGLY_WEAKENING

Checkpoint	Ref \$	Price \$	Episode move	Peak move	Giveback	State
RE-ACCEL 13:55	103.00	110.10	+6.9%	+8.2%	~16%	RE_ACCELERATING
CLOSE 16:00	103.00	107.40	+4.3%	+8.2%	~48%	CLOSED (silent)

9.6 Alternate afternoon: true REVERSAL (teaching case)

Suppose instead of holding into the close, SNDK sold off hard after 14:30: price falls through the \$103 reference and the 5m / 15m windows print $\leq -3\%$. Then:

- originalMoveErased(UP) becomes true (price $\leq$ reference)
- oppositeThresholdHit finds DOWN windows above thr
- Emit MOMENTUM_REVERSED (push) + end episode with REVERSAL
- FULL re-arm: UP direction disarmed until all UP windows cool below thr-1pp

A fade to \$105 with reference \$103 is still an UP episode (weakening) — **not** a reversal. This distinction is intentional so we do not cry “reversed” on every healthy pullback.

9.7 What the dashboard shows alongside the episode

While the episode is ACTIVE, the returns grid still paints every window (1m, 5m, ... day, multi-day). Cards above threshold highlight; multi-day cards never create a second episode. Operators can retune thresholds per asset class without redeploying code. Mobile detail payloads include frozen identity, peak/trough, giveback, and notification history for that episodeId.

9.8 Worked START micro-example (math)

At 10:18 ET, latest price \$108.50. Candle ~5 minutes earlier $\approx$ \$103.00.

move = $(108.50 - 103.00) / 103.00 \times 100 = +5.34\%$.

5m threshold = 3% $\rightarrow$ hit. 15m might be +4.8% (also a hit) but longer rank loses. day = $(108.50 - 100) / 100 = +8.5\%$ (also a hit) but day ranks like a full session, so 5m still wins. Fresh-cross filter ensures 5m was not already above thr last poll.

9.9 Worked ACCEL micro-example (math)

After START, lastNotifiedEpisodeMovePct = +5.34%. At 11:20 price = \$111.40, reference still \$103.00.

episode move = $(111.40 - 103.00) / 103.00 \times 100 = +8.16\%$.

Extension vs last notify = $8.16 - 5.34 = +2.82 \text{ pp} \geq 2.0 \rightarrow$ ACCELERATING push fires. lastNotifiedEpisodeMovePct becomes +8.16%.

10. Daily Digest & related bulletins

Two related but **separate** systems sit next to episodes. They must not be confused with episode START/ACCEL logic.

Daily Digest

Slot	When (US RTH defaults)	Role
OPEN	Regular open + 10 minutes	Early session snapshot for equities/indexes on the watchlist
MIDDAY	Midpoint of regular session	Mid-session check-in
CLOSE	Regular close + 5 minutes	End-of-day summary

Digests are deduped per (ticker, session_date, slot), skip US market holidays, and only run for equities/indexes. They do **not** create, modify, expire, or reset episodes. Disable with MOMENTUM_DAILY_DIGEST_ENABLED=0.

Market session bulletins

Optional session-boundary messaging (feature-flagged). Also independent of the episode state machine. Both digests and episode alerts can share the Expo delivery path and watchlist gating.

11. Configuration reference & glossary

Key defaults (equity / episode policy)

Parameter	Default	Role
Poll interval	60s	How often each ticker is evaluated
Accel alert delta	2.0 pp	Extra episode move needed for ACCEL push
Material progress	0.5 pp	Resets inactivity clock
Inactivity expiry	60 min	Silent EXPIRED without progress
Giveback → WEAKENING	25%	State only (silent)
Hysteresis back to HOLDING	20%	Avoids state flicker
Strong weakening	60%	Push once per fade cycle
Re-arm buffer	1.0 pp	Cool below thr-buffer to re-arm direction
START push max age	5 min	Drop late START pushes
Day threshold	5%	Vs previous close
5m / 10m / 15m thr	3%	Short windows
Max watched tickers	20	Background loop cap

Glossary

Term	Definition
Episode	One directional momentum case file for a ticker
Reference	Frozen anchor price from the START window
Threshold hit	$ \text{return} \geq \text{configured absolute \% for that window}$
Fresh cross	Hit that was not already above thr last poll
Giveback	Fraction of peak directional move surrendered
pp	Percentage points (difference of percents), not relative %
FULL re-arm	All same-dir windows must cool before new START that way
SESSION re-arm	Market-close end; next day uses crossing filter only
Blink-only window	Shown in UI; cannot start/keep episodes (e.g. 1w)
Idempotency key	Prevents duplicate pushes for the same logical event

12. Product principles & what “good” looks like

- **Narrative over noise** — one episode story beats ten overlapping window pings.
- **Shortest window authenticity** — START copy reflects the tightest real impulse, not a stale long lookback.
- **Push only on material change** — +2pp, 60% fade, re-accel, reverse; hold/weaken quietly.
- **Fade ≠ reverse** — language and events keep “giving back” separate from “thesis flipped.”
- **Session hygiene for stocks** — do not drag an intraday equity episode across the overnight gap as if nothing changed.
- **Operator control** — thresholds per asset class, feature flags, dry-run / allowlist for safe testing.
- **Explainability** — frozen reference, exact minutes, peak, giveback, and event history make every push auditable.

Suggested talking points for a leadership review

Question they may ask	Answer
Will this spam users?	No — selective pushes; HOLDING/WEAKENING/expiry/close are silent.
Why only one episode?	One coherent story per ticker; accel extends the same case.
Why shortest window?	Most timely impulse; avoids “24h was already up” false drama.
What about overnight?	Equities archive at cash close; next day needs a fresh cross.
Is the SNDK day real prints?	Section 9 is an illustrative walkthrough of production rules.
How is this different from digests?	Digests are scheduled session summaries; episodes are event-driven.

Document status. Describes the V1 Trigger Episode Alert logic as implemented in the NewsLabs momentum engine (server/momentum/*). Thresholds and timers are defaults; production values may be tuned in settings without changing the story model. For engineering detail, see unit tests under server/momentum/*.test.js and the episode state machine in episode.js.

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